

[Dashboard](#)[my courses](#)[AMCN SS 2022](#)[wireless transmission](#)[Quiz: Wireless Transmission - Frequency and Signals](#)

Started at	Sunday, April 10, 2022, 3:58 p.m
status	Completed
Finished at	Sunday, April 10, 2022 at 11:56 p.m
Spent time	7 hours 58 minutes
valuation	10.00 out of 10.00 (100 %)
feedback	Congratulations, excellent performance (90-100%).

question **1**

Complete

Score 1.00 out
of 1.00

How does the frequency spectrum of a sinusoidal signal with 100 MHz look like?

Choose one or more answers:

- ☒ a. There is a single peak at 100MHz only
- ☐ b. There are peaks at 100, 200, 400, 800, 1600 ... MHz
- ☐ c. There are peaks at 100, 200, 300, 400, ... MHz
- ☐ i.e. There are peaks at 100, 300, 500, 700, ... MHz

question **2**

Complete

Score 1.00 out
of 1.00

How does the frequency spectrum of a symmetric digital signal with a frequency of 100 MHz look like?

Choose one or more answers:

- ☐ a. There is a single peak at 100MHz only
- ☐ b. There are peaks at 100, 200, 300, 400, ... MHz
- ☒ c. There are peaks at 100, 300, 500, 700, ... MHz
- ☐ i.e. There are peaks at 100, 200, 400, 800, 1600 ... MHz

question **3**

Complete

Score 1.00 out
of 1.00

How does the frequency spectrum of a symmetric triangular signal with a frequency of 100 MHz look like?

Choose one or more answers:

- ☒ a. There are peaks at 100, 300, 500, 700, ... MHz
- ☐ b. There are peaks at 100, 200, 300, 400, ... MHz
- ☐ c. There are peaks at 100, 200, 400, 800, 1600 ... MHz
- ☐ i.e. There is a single peak at 100MHz only

question **4**

Complete

Score 1.00 out
of 1.00

What are the pros and cons of using higher frequencies for communications?

Choose one or more answers:

- ☒ a. Problems with penetration of obstacles as walls, doors, windows, etc.
- ☐ b. More unused frequency space available
- ☐ c. Hardware technology is simpler
- ☒ i.e. Smaller antennas

question **5**

Complete

Score 1.00 out
of 1.00

What is the wavelength of the GSM 1800 MHz band?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. 66,6 m
- ☐ b. 33,3 m
- ☐ c. 10 cm
- ☐ d. 1 cm
- ☒ e. 16,7 cm
- ☐ f. 3,3 cm

Frage **6**

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the wavelength of the GSM 900 MHz band?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. 33,3 cm
- ☐ b. 3,3 cm
- ☐ c. 3,3 m
- ☐ d. 1 m
- ☐ e. 10 cm



Frage 7

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the wavelength of WLAN at 2,4 band?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. 25 cm
- ☒ b. 12,5 cm
- ☐ c. 8m
- ☐ d. 6,25cm

Frage 8

Vollständig

Erreichte
Punkte 1,00 von
1,00

What is the wavelength of WLAN at 5 GHz band?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. 3 cm
- ☒ b. 6 cm
- ☐ c. 18m
- ☐ d. 12 cm



Frage 9

Vollständig

Erreichte
Punkte 1,00 von
1,00

Which GSM frequency band typically provides better indoor coverage (from outdoor base stations)? What is the justification for this?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Higher frequencies are easily absorbed by walls
- ☐ b. Lower frequencies have problems with smaller rooms
- ☒ c. GSM 900
- ☐ d. GSM 1800
- ☐ e. Larger wavelengths are easily absorbed by walls

Frage 10

Vollständig

Erreichte
Punkte 1,00 von
1,00

Which of the following statements hold for the OSI model?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Each layer provides a service interface called service access point to its lower layer
- ☒ b. It divides functions needed for communications in different layers
- ☐ c. The functionality implemented by a layer is called a service
- ☒ d. It defines an architecture to structure communication systems
- ☒ e. The layers build on each other

Vorhergehende
Aktivität

Nächste Aktivität

◀ Shannon Capacity and
Modulation (Video)

Direkt zu:

Quiz: Drahtlose
Übertragung -
Signalausbreitung ▶



[imprint](#)

How to reach us:

 <https://www.tu-ilmenau.de>

 [+49 3677 69-1111](tel:+493677691111)

 1111@tu-ilmenau.de